The Lifestyle Sleep Connection: Insights From Behavioral Data Mining

Joshua Alcantara, James Alba, Jasmine Saunders

College of Communication & Information, Florida State University

LIS 4671: Data Mining and Analytics

Dr. Subhasree Sengupta

December 12, 2025

PROJECT DESCRIPTION

This project investigates the relationship between lifestyle behaviors and sleep outcomes using the Sleep Health and Lifestyle dataset (Tharmalingam, 2023). Sleep quality is known to be connected to physical and mental health, however the behavioral patterns that actually influence sleep are somewhat difficult to quantify without any structured analytical methods. By applying data mining techniques, our goal is to identify meaningful trends within the dataset, uncover relationships amongst the variables, and explore how everyday habits might influence things like sleep duration, sleep quality, and even the occurrence of sleep disorders.

The dataset includes 400 individuals and has a range of demographic and lifestyle features like age, occupation, physical activity level, stress level, heart rate, daily step counts, sleep duration, sleep quality, and sleep disorder status. Since these variables exist in both behavioral and physiological domains, they give us a solid foundation for things like clustering, pattern recognition, and predictive modeling.

This project aligns with the broader goals of data mining by combining methods of structured exploratory analysis with supervised and unsupervised learning techniques. Exploratory data analysis, or EDA, allows us to look further into variability in the dataset, detect any initial associations, and evaluate data quality; while predictive modeling allows us to evaluate potential risk indicators for sleep disorders. Through these methods, we aimed to develop a clearer understanding of how stress, activity levels, and other lifestyle factors influence sleep.

Overall, this project gives insights into an important question in public and personal health, **What everyday behaviors meaningfully affect sleep, and how can data-driven analysis improve our understanding of these patterns?** Our findings have potential value for both individuals looking to improve their well being and public health initiatives focused on sleep-related health risks.

OVERVIEW OF OBJECTIVES

Key Questions We Wanted to Address

We selected this topic because sleep is a large factor in one's overall health, yet many people struggle with poor sleep quality due to things like stress, inactivity, or other lifestyle related factors.

The key questions that guided this project were focused on understanding how everyday lifestyle behaviors influence sleep outcomes. Specifically, we aimed to identify which behavioral and physiological factors were most strongly associated with sleep quality and sleep duration, and whether stress level emerged as a major predictor of poor sleep. We also attempted to examine whether individuals with sleep disorders, like insomnia or sleep apnea, displayed behavioral patterns different to those without a disorder. We also explored whether supervised learning methods could accurately classify individuals at risk for a sleep disorder based on metrics like sleep duration, physical activity, and stress. Finally, we considered what natural groupings or behavioral profiles would present themselves through unsupervised learning techniques. Overall, these questions shaped the direction of our EDA and informed the modeling approaches that we used in the later stages of the project.

ANALYTICAL OVERVIEW

Our analytical process had three main stages: exploratory data analysis (EDA), supervised learning, and evaluation and interpretation of findings. The EDA phase focused on understanding the dataset's structure and basic patterns. This included examining distributions, checking for data quality issues, and exploring initial relationships using visualizations.

Building on insights from EDA, we applied a supervised learning model to test whether sleep quality could be predicted from lifestyle and demographic variables. Specifically, we trained a logistic regression classifier to distinguish between individuals with good versus poor self-reported sleep quality using features such as sleep duration, stress level, physical activity, daily steps, and sleep disorder status. Finally, we evaluated the model with standard classification metrics and interpreted the results in light of our original research questions. This workflow provided both descriptive insight (what patterns exist in the data) and predictive insight (how well those patterns can be used to identify individuals at risk of poor sleep).

EXPLORATORY ANALYSIS

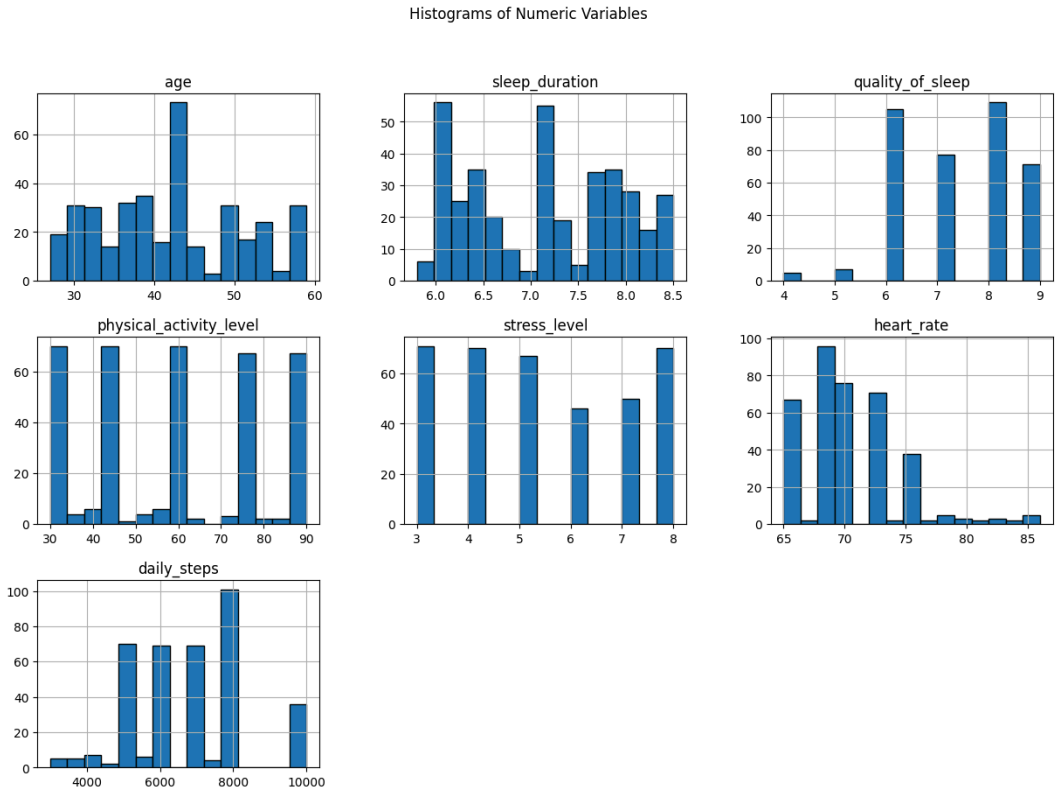
Data Quality and Structure

Initial examination showed 400 observations and no missing values across the sleep and lifestyle variables. Column names were standardized to a consistent format and descriptive statistics confirmed that all numeric values fell within realistic human ranges. Daily step counts exhibited a wide spread, while sleep duration, sleep quality, and heart rate showed narrower distributions. Boxplots revealed a small number of expected outliers, primarily among daily step values, which were retained because they represent realistic variability in daily activity.

Univariate Analysis

Histograms were used to look at the distribution of numeric variables. Sleep duration was centered around 6-8 hours, while we saw sleep quality clustered around moderate to moderately high values (6-8). Stress level showed a broad distribution across the 1-10 scale, which helped show later correlations. Physical activity level and daily steps also showed a spread, suggesting diverse lifestyles within the sample.

Figure 1: Histogram Grid of Numeric Variables

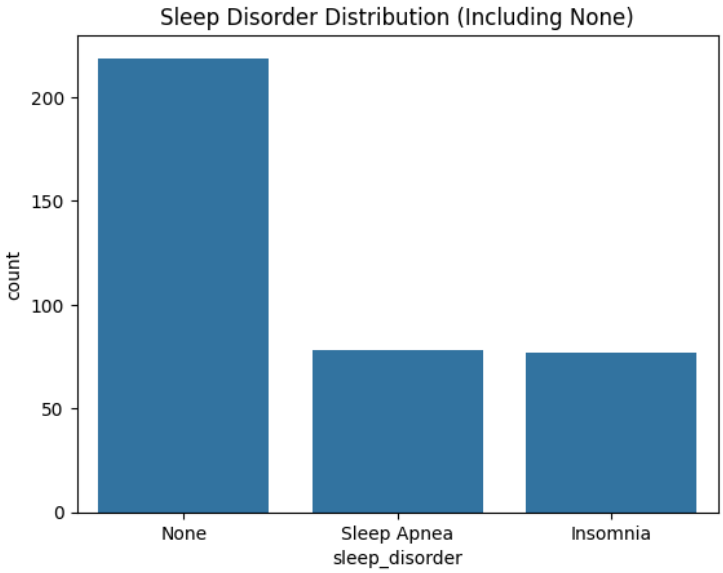


Distribution of core behavioral and physiological variables in the dataset.

Categorical Variable Exploration

The sleep_disorder variable contained three categories: Insomnia, Sleep Apnea, and None. Frequency counts showed that a large proportion of individuals reported no sleep disorder, while the occurrences of Insomnia and Sleep Apnea were nearly equal. Gender distribution was similarly balanced, and occupations existed in a broad range.

Figure 2: Bar Chart of Sleep Disorder Categories

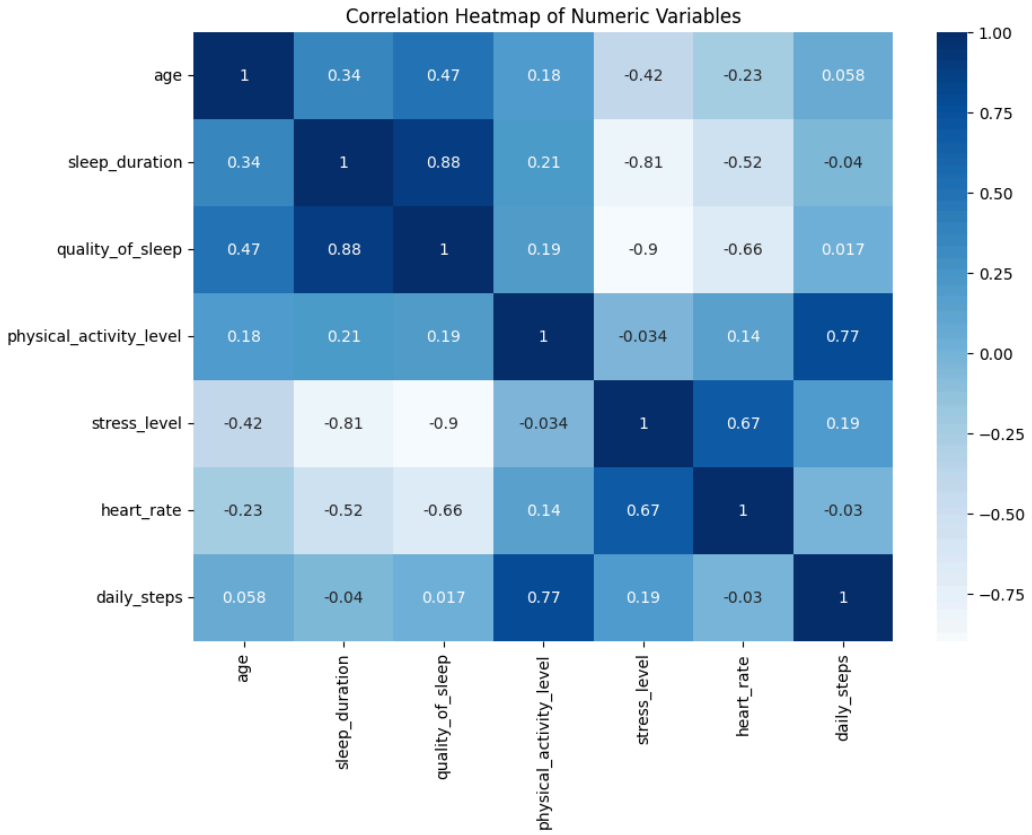


Frequency of reported sleep disorders (Insomnia, Sleep Apnea, None).

Bivariate and Correlation Analysis

Scatterplots and correlation heatmaps were generated to look at relationships between behavioral metrics and sleep outcomes. Stress level showed a visible downward trend when plotted against sleep duration, suggesting that higher stress was linked to reduced sleep. The heatmap further quantified these findings. Sleep duration and sleep quality demonstrated a very strong positive correlation ($r \approx 0.88$). Stress level showed extremely strong negative correlations with both sleep duration ($r \approx -0.81$) and sleep quality ($r \approx -0.90$). Heart rate was moderately associated with stress and sleep quality, while physical activity level had a strong relationship with daily steps but only a weak relationship with sleep-related variables

Figure 3: Correlation Heatmap of Numeric Variables



Strength and direction of relationships among key lifestyle and sleep variables.

SUPERVISED LEARNING MODEL

Modeling Objective

To address whether we can predict good versus poor sleep quality from lifestyle factors we trained a supervised classification model. We defined a binary target variable, good_sleep, where values of 7 or higher on the 0-10 sleep quality scale were labeled as “good sleep” (1) and values below 7 were labeled as “poor sleep” (0). The goal of the model was to estimate the probability that an individual reports good sleep quality based on their demographic and behavioral characteristics.

Model Architecture and Variables

We used logistic regression, a supervised learning method commonly used for binary classification. The predictor variables included both continuous and categorical features. Continuous variables were: age, sleep duration (hours), physical activity level, daily step count, stress level (1-10 scale), and resting heart rate. Categorical variables were: gender, occupation, BMI category, and sleep disorder status (none, insomnia, sleep apnea).

Before training the model, we standardized the continuous variables to have mean 0 and standard deviation 1 so that variables measured on larger scales (daily steps) would not dominate the model. Categorical variables were transformed using one-hot encoding, which creates binary indicator columns for each category. We then split the data into training (80%) and testing (20%) sets using stratified sampling to preserve the proportion of good vs. poor sleepers. All preprocessing steps and the logistic regression model were combined into a single scikit-learn pipeline to ensure the same transformations were applied to both training and test data.

Evaluation Metrics and Performance

The trained model performed well on the test set. Overall accuracy was approximately 0.97, meaning the model correctly classified about 97% of the 31 test cases. The detailed classification report showed balanced performance across both classes. For the *poor sleep* class (0), precision was 0.94 and recall was 1.00, indicating that when the model predicted poor sleep it was usually correct, and it successfully identified all true poor sleepers in the test set. For the *good sleep* class (1), precision was 1.00 and recall was 0.94, so every predicted good sleeper in the test set actually reported good sleep, and the model missed only a small fraction of true good sleepers. Macro and weighted F1-scores were both 0.97, confirming consistently strong performance across classes.

The confusion matrix (Figure 4) shows that there was only a single misclassified case out of 31, where an individual who reported good sleep was predicted as a poor sleeper. All other observations fell on the diagonal of the matrix, reflecting a low error rate and no systematic bias toward one class.

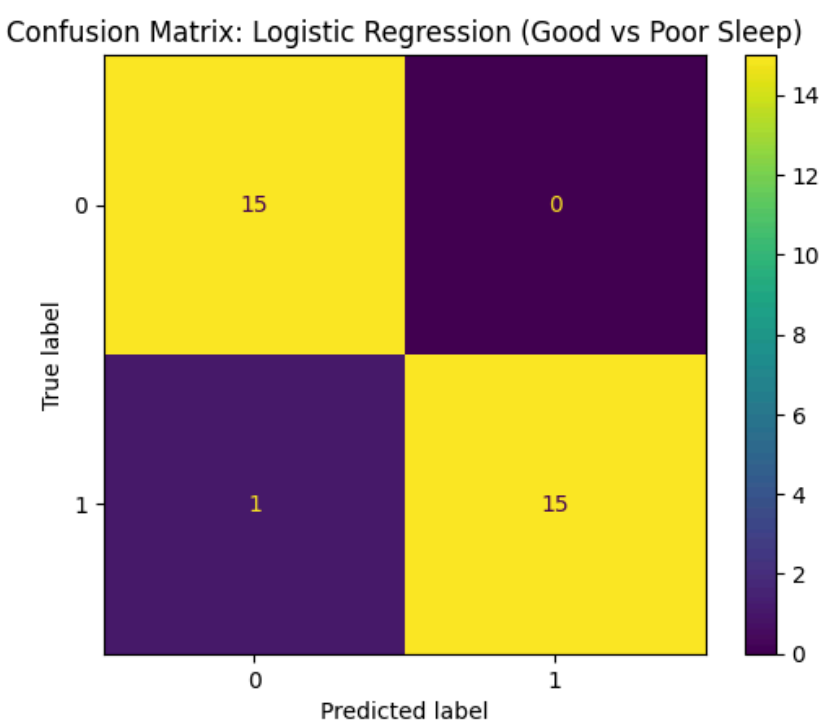


Figure 4

Interpretation of Results

The logistic regression results are consistent with patterns observed in the exploratory analysis and help clarify how each variable influences the prediction. In this model, each learned coefficient is a weight that shows how changing a feature affects the predicted probability of good sleep quality while holding other variables constant. A positive coefficient means that higher values of that variable increase the predicted probability of good sleep; a negative coefficient means that higher values decrease it. Larger coefficients in absolute value indicate variables that have a stronger impact on the prediction.

In our model, longer sleep duration and higher physical activity levels had positive coefficients, indicating that sleeping more hours and being more active are associated with a higher probability of reporting good sleep quality. Stress level had a strong negative coefficient, meaning that as stress increases, the model's predicted probability of good sleep drops sharply. Indicators for having a sleep disorder (insomnia or sleep apnea) were also negative, suggesting that, all else equal, individuals with these conditions are less likely to be classified as good sleepers.

These patterns closely match what we observed in the exploratory analysis, where sleep duration and sleep quality were highly positively correlated and stress showed strong negative relationships with both sleep duration and sleep quality. The supervised model therefore reinforces the earlier descriptive findings: stress, sleep duration, and activity level are not only correlated with sleep outcomes but are also the most influential predictors when the model decides whether someone is likely to experience good or poor sleep in this dataset.

REFLECTION AND HUMAN-CENTERED CONSIDERATIONS

Human-Centered Insights

Our analysis suggests that sleep health in this sample is strongly shaped by a small set of lifestyle factors, especially stress level, sleep duration, and physical activity. From a human-centered perspective, this means that interventions aimed at improving sleep quality, rather than only on demographic factors that people cannot easily change. The clear separation between individuals with and without reported sleep disorders in our model also highlights that people with chronic conditions may need more targeted support, combining lifestyle counseling with clinical evaluation.

Model Limitations

Despite its strong performance, this model has several limitations. The dataset is relatively small (400 individuals), and the test set contained only 31 observations, so even a single misclassification has a noticeable impact on the metrics. All variables are cross-sectional and self-reported, which may introduce measurement error or bias. Finally, logistic regression captures associations rather than causality: while stress and activity are strong predictors of sleep quality, we cannot conclude that changing any one factor will directly cause better or worse sleep. These limitations are important to keep in mind when interpreting the model's predictions and considering their relevance in real-world settings.

Teamwork Distribution

Joshua Alcantara completed the Exploratory Data Analysis (EDA) phase of the project, establishing the foundation for the modeling stages. Work included importing and preparing the dataset, performing data quality checks, identifying missing values, and ensuring variables were properly cleaned and standardized. Descriptive statistics and initial visualizations were generated. Also contributed to drafting written components including project description, overview of objectives, and the exploratory analysis sections.

Jasmine Saunders led the modeling and interpretation components, including building and comparing supervised and unsupervised models, selecting the final modeling approach, and generating key visualizations for the results. She also drafted the results, discussion, and implications sections, integrating statistical findings with substantive insights about sleep health and lifestyle.

James Alba focused on feature engineering and model evaluation, refining variable selection, tuning model parameters, and assessing performance using accuracy, precision, recall, and related metrics. He also coordinated consolidation of code, screenshots, and figures, and contributed to editing the full report and preparing materials for the final presentation.

References

Tharmalingam, L. (2023). *Sleep health and lifestyle dataset*. Kaggle. https://www.kaggle.com/datasets/uom190346a/sleep-health-and-lifestyle-dataset